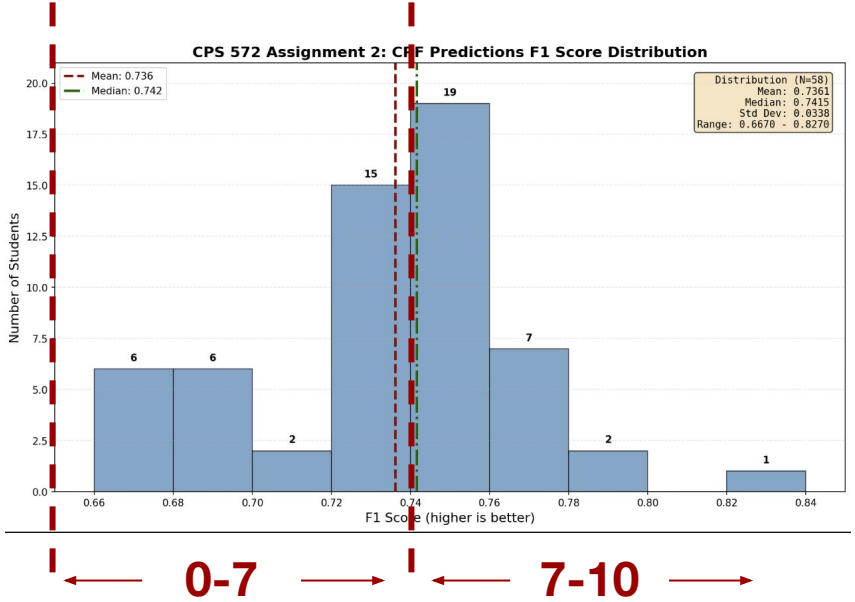
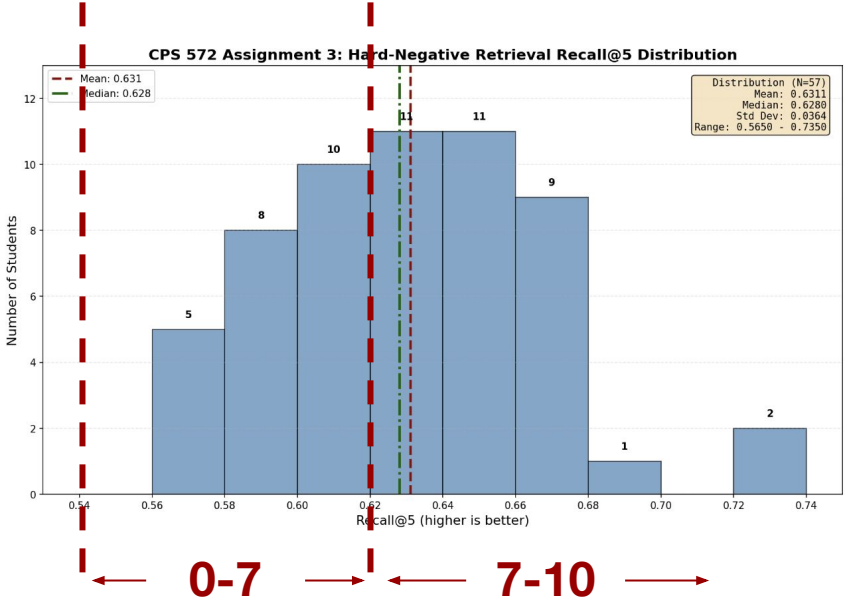
COMPSCI 572.01 (NLP)
Interpretability & Ethics

A2 and A3 grades out





A2

What worked:

- Linear-chain CRFs
  - Better features: prefix, suffix, capitalization
- Neural CRFs (BiLSTM+CRF)
  - Character-level models (generalization to unseen entities)
  - Pretrained word vectors (Glove)
  - Checkpoint selection using F1 rather than loss

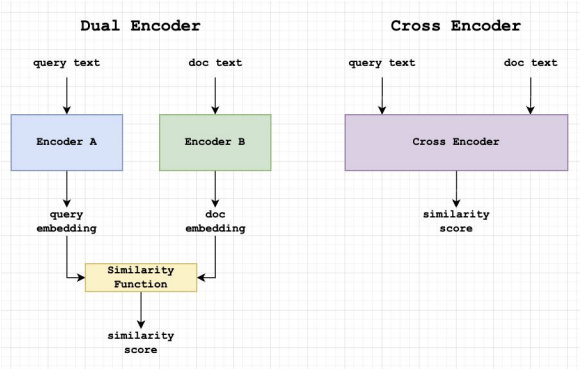
What didn’t work:

- Bad features (sentence position, sequence length)
- Transformers (limited data)
- No pretrained word embeddings, no features in neural models
- Reports without error analysis or ablations

A3

What worked:

- Cross-encoder reranking on top of bi-encoder retrieval
- Iterative hard-negative mining
- Sparse-dense combined approaches
- Architectural details: uncased BERT and capitalized text



Overview

- Interpretable vs Black box models
- Global interpretations (aka probing)
- Local interpretations
  - LIME, Influence functions
- Ethical considerations in NLP

Duke Course Evaluations
https://eval-duke.evaluationkit.com/

Why interpretation?

Model interpretation refers to techniques which aim to derive an understanding of the model's behavior in making predictions

Why should we care about this?

- Debugging
- Increasing trust
- Complying with regulations
- Ensuring fairness
- ...

Interpretable vs Black Box Models

- Black boxes: prediction function is too complicated or proprietary
  - E.g. neural networks
- Interpretable models: prediction function is understandable by humans

Interpretability vs Explainability in ML *Cynthia Rudin*

Interpretable vs Black Box Models

- Black boxes: prediction function is too complicated or proprietary
  - E.g. neural networks
- Interpretable models: prediction function is understandable by humans
  - E.g. linear models

$$f(\mathbf{x}) = \overset{2.0}{w_1}\mathbf{x}_1 + \overset{0.01}{w_2}\mathbf{x}_2$$

Estimates Rent

House

Area

Population Density

Interpretability vs Explainability in ML *Cynthia Rudin*

Explaining Black Boxes

global interpretation

- What is the model learning?

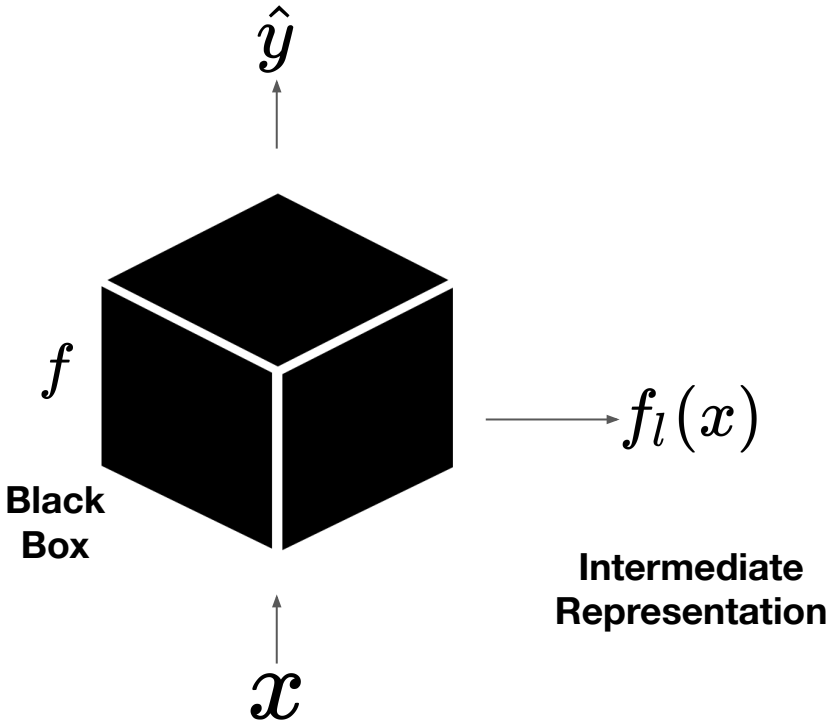
- Can we explain the prediction?

local interpretation

Figure credit: Danish Pruthi's slides

Global interpretations

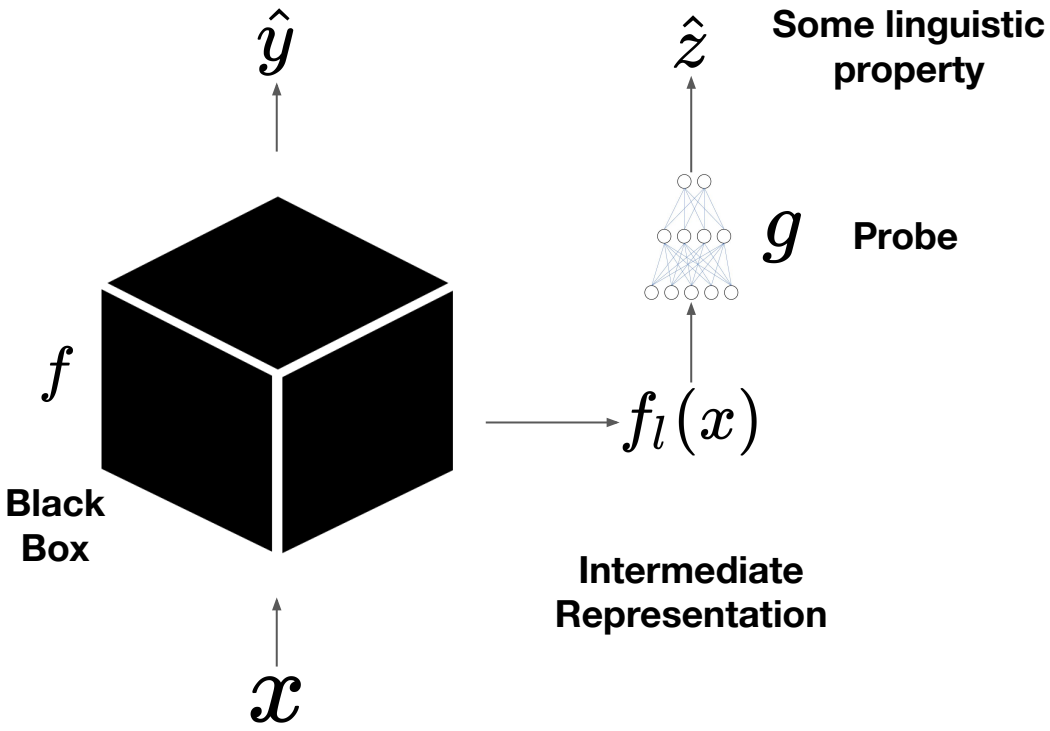
- **Probing**: predict linguistic properties from model representations



Probing Classifiers *Belinkov 2021*

Global interpretations

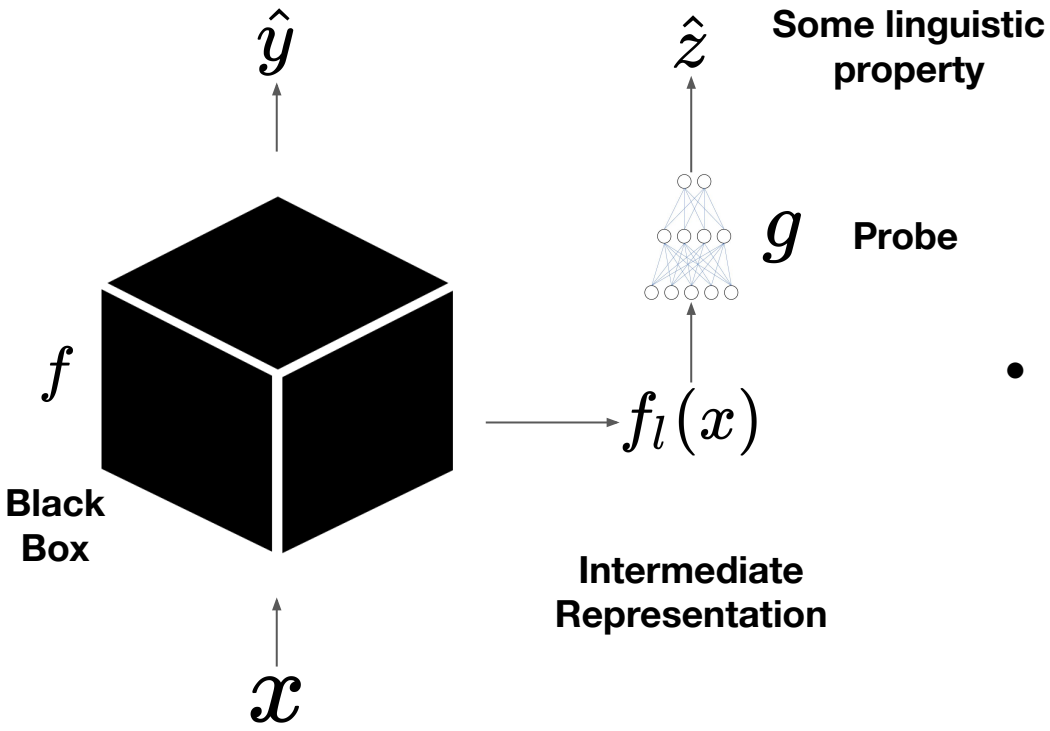
- **Probing**: predict linguistic properties from model representations



Probing Classifiers *Belinkov 2021*

Global interpretations

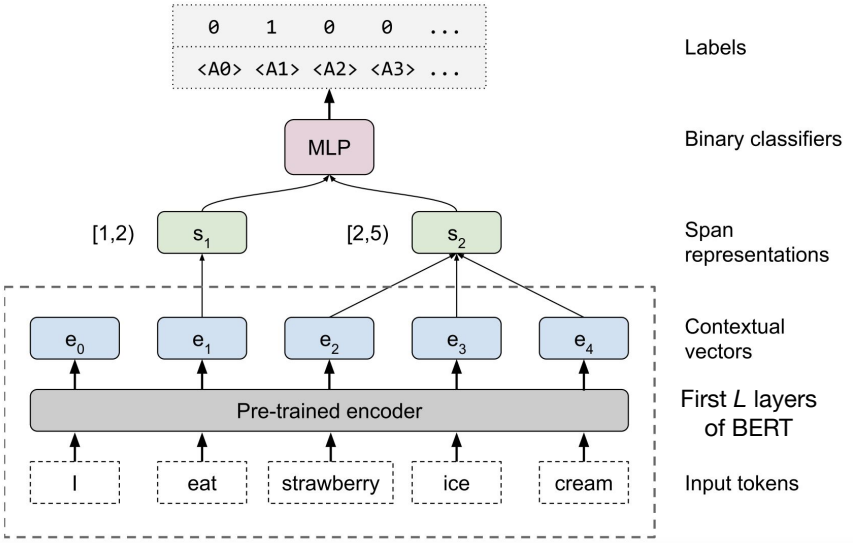
- **Probing**: predict linguistic properties from model representations

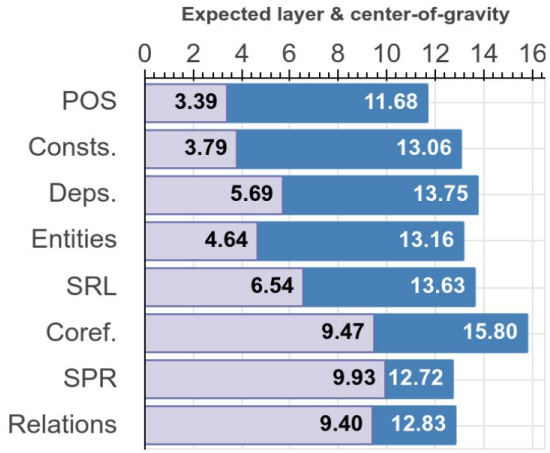


- If *g* has high accuracy => linguistic property encoded in the representation

Probing Classifiers *Belinkov 2021*

Probing BERT



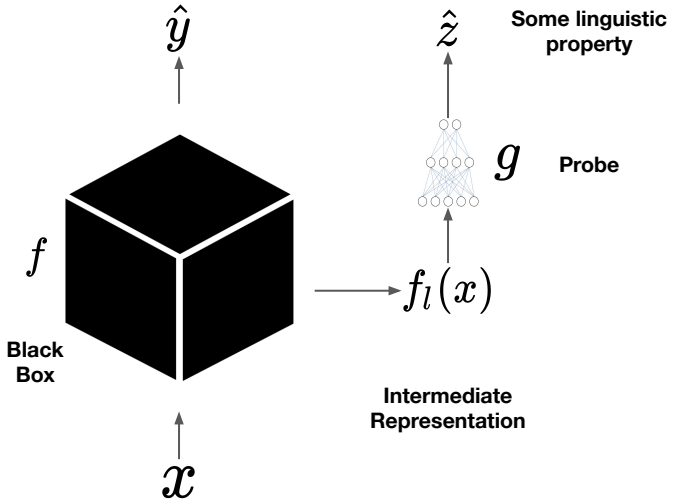


- Lower layers capture syntactic information
- Higher layers capture semantic information

BERT Rediscovers the Classical NLP Pipeline *Tenney et al, 2019*

Limitations

- Is the property encoded in the representation or learnt by *g*?
- Even if the property is encoded in the representation, is it used for making predictions?



Hewitt et al, 2019

Explaining Black Boxes

global interpretation

- What is the model learning?

- Can we explain the prediction?

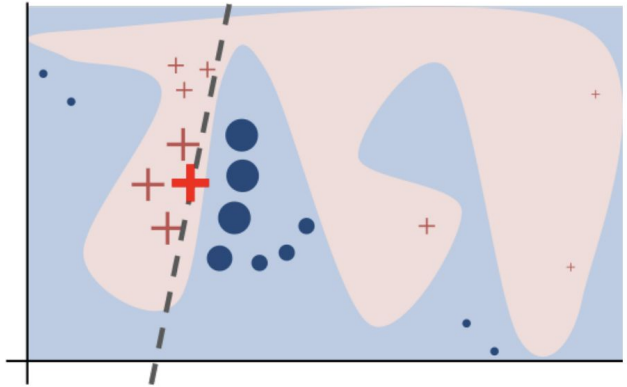
local interpretation

Can we identify factors which are responsible for a specific output?

Figure credit: Danish Pruthi's slides

LIME (Locally interpretable model-agnostic explanations)

The movie is mediocre, maybe even bad.	**Negative** 99.8%
The movie is mediocre, maybe even ~~bad~~.	**Negative** 98.0%
The movie is ~~mediocre~~, maybe even bad.	**Negative** 98.7%
The movie is ~~mediocre~~, maybe even ~~bad~~.	**Positive** 63.4%
The movie is ~~mediocre~~, ~~maybe~~ even ~~bad~~.	**Positive** 74.5%
The ~~movie~~ is mediocre, maybe even ~~bad~~.	**Negative** 97.9%



The movie is mediocre, maybe even bad.

[Explaining the Predictions of Any Classifier](#) *Ribeiro et al, 2016*

Influence functions

- Can we identify training instances which maximally influence a test instance?

Understanding Black-box Predictions *Koh et al, 2017*

Influence functions

- Related: how would the parameters change if we adjust a training instance?

$z_1, \ldots, z_n \quad z_i = (x_i, y_i)$ ← Training data

$\hat{\theta} = \text{argmin}_\theta \frac{1}{n} \sum_{i=1}^{n} L(z_i, \theta)$

Trained parameters

Loss on one instance

Understanding Black-box Predictions *Koh et al, 2017*

Influence functions

- Related: how would the parameters change if we adjust a training instance?

$$z_1, \ldots, z_n \quad z_i = (x_i, y_i)$$

$$\hat{\theta} = \text{argmin}_\theta \frac{1}{n} \sum_{i=1}^{n} L(z_i, \theta)$$

$$\hat{\theta}_{\epsilon,z} = \text{argmin}_\theta \frac{1}{n} \sum_{i=1}^{n} L(z_i, \theta) + \epsilon L(z, \theta)$$

Trained parameters if we **up-weight *z***

$$\mathcal{I}_{\text{up,params}}(z) = \frac{d\hat{\theta}_{\epsilon,z}}{d\epsilon}\Big|_{\epsilon=0}$$

Influence of *z* on parameters

Understanding Black-box Predictions *Koh et al, 2017*

Influence functions

- Related: how would the parameters change if we adjust a training instance?

$$z_1, \ldots, z_n \quad z_i = (x_i, y_i)$$

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$$\left\{ H_{\hat{\theta}} = \frac{1}{n} \sum_{i=1}^n \nabla_\theta^2 L(z_i, \hat{\theta}) \right\}$$

$$\mathcal{I}_{\text{up,params}}(z) = \frac{d\hat{\theta}_{\epsilon,z}}{d\epsilon}\Big|_{\epsilon=0} \approx -H_{\hat{\theta}}^{-1} \nabla_\theta L(z, \hat{\theta})$$

Hessian matrix

Gradient of parameters

Understanding Black-box Predictions *Koh et al, 2017*

Influence functions

- Chain rule for determining influence of training instance on test instance

$$z_1, \ldots, z_n \quad z_i = (x_i, y_i)$$

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$$\mathcal{I}_{\text{up,loss}}(z, z_{\text{test}}) = \frac{dL(z_{\text{test}}, \hat{\theta}_{\epsilon,z})}{d\epsilon}$$

Influence of *z* on *z_test*

Understanding Black-box Predictions *Koh et al, 2017*

Influence functions

- Chain rule for determining influence of training instance on test instance

$$z_1, \ldots, z_n \quad z_i = (x_i, y_i)$$

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$$\mathcal{I}_{\text{up,params}}(z) = \frac{d\hat{\theta}_{\epsilon,z}}{d\epsilon}\Big|_{\epsilon=0} \approx -H_{\hat{\theta}}^{-1} \nabla_\theta L(z, \hat{\theta})$$

$$\mathcal{I}_{\text{up,loss}}(z, z_{\text{test}}) = \frac{dL(z_{\text{test}}, \hat{\theta}_{\epsilon,z})}{d\epsilon} = \nabla_\theta L(z_{\text{test}}, \hat{\theta})^T \frac{d\hat{\theta}_{\epsilon,z}}{d\epsilon}\Big|_{\epsilon=0} = \boxed{-\nabla_\theta L(z_{\text{test}}, \hat{\theta})^T H_{\hat{\theta}}^{-1} \nabla_\theta L(z, \hat{\theta})}$$

Understanding Black-box Predictions *Koh et al, 2017*

Overview

- Interpretable vs Black box models
- Global interpretations (aka probing)
- Local interpretations
  - Attention, LIME, Influence functions
- **Ethical considerations in NLP**

NLP and people

The common misconception is that language has to do with *words* and what they mean.

It doesn't.

It has to do with ***people*** and what *they* mean.

Decisions we make about our data, methods, and tools are tied up with their impact on people and societies.

Slide credit: Yulia Tsvetkov

Necessary questions about any ML system

1. Is it ethical to build such a system?
2. Who benefits from it?
3. Who might be harmed by it?
4. How are these benefits and harms distributed?
5. Who is responsible?

These questions also apply to any data collected!

Case study: the “AI Gaydar” system

- A real 2017 paper written by real people from Stanford (business school).
  - (To be fair, the paper sort of presents their findings as a warning).
- Goal: identify sexual orientation of men or women from face pictures.
  - Many similar recent systems (identifying criminals, terrorists, etc).
- Data: ~35K images of white people from an American dating site.
  - Sexual orientation label based on how people self-identified on the site.
  - Proportion of men/women and gay/straight roughly balanced.
- Model: neural net features + logistic regression.

Slide inspiration: Yulia Tsvetkov

Everything about this is bad

Bad from a technical perspective too, but the ethical issues are most important.

- Who benefits from such a system?

Everything about this is bad

Bad from a technical perspective too, but the ethical issues are most important.

- Who benefits from such a system?
- Who is harmed by it?

Everything about this is bad

Bad from a technical perspective too, but the ethical issues are most important.

- Who benefits from such a system?
- Who is harmed by it?
  - Anyone predicted to be gay. Can be used to discriminate or worse.

What about the data?

The photos were downloaded from a dating site. But:

- Legal ≠ Ethical
- Public ≠ Publicized
- No explicit consent from people in the photos.

Slide credit: Yulia Tsvetkov

What about the data?

The photos were downloaded from a dating site. But:

- Legal ≠ Ethical
- Public ≠ Publicized
- No explicit consent from people in the photos.

The data is clearly biased:

- Only white people, self-identified sexual orientation, each label uniformly distributed, dating profile pics.

Slide credit: Yulia Tsvetkov

Uninterpretable models exacerbate things

- Was the model just capturing spurious correlations?
- How will the model perform if we change the input?

The pervasiveness of bias

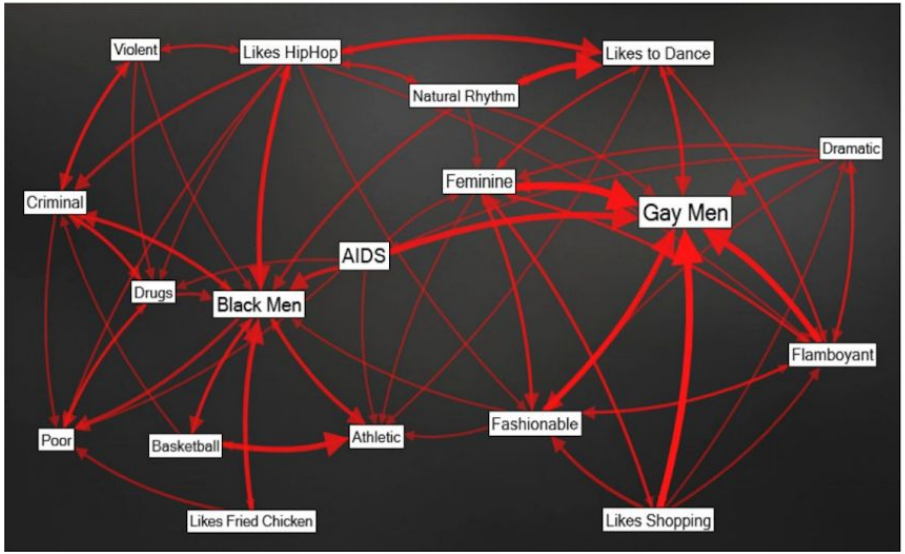


Image credit: Geoff Kaufman

Implicit biases are pervasive, unconscious, and can automatically influence the ways in which we see and treat others, even when we are determined to be fair and objective.

Slide credit: Yulia Tsvetkov

The pervasiveness of bias

- **An important consequence:** the data we train our models on is biased.
- Models will therefore reflect, and often even amplify, this bias.

Other things to worry about

(Environmental) costs of training models:

- Frontier models take >$100M to train
- Inference costs are also significant
- BERT-base pretraining carbon emissions are roughly equivalent to being a passenger on a flight from NY to SF (Strubell et al., 2019).
- Also: how are the benefits/harms distributed? (Bender et al., 2021)

Slide credit: Greg Durrett

Recap: for every dataset/system, need to ask

1. Is it ethical to build such a system?
2. Who benefits from it?
3. Who might be harmed by it?
4. How are these benefits and harms distributed?
5. Who is responsible?

We've covered a lot of ground!

1. **(Classical) NLP**
   - **Text classification** (generative, discriminative, neural)
   - **Language modeling** (n-gram models, perplexity, information theory, smoothing, log-linear models, feed-forward models, RNNs, attention, transformers)
   - **Structured prediction** (generative and discriminative models of sequences, tractable and intractable inference, neural extensions)
   - **Information Retrieval** (TF-IDF, Dual Encoders, word2vec)
2. **Large Language Models**
   - **Training and Scaling** (pretraining, post-training, scaling laws, mixture-of-experts)
   - **Efficiency** (LoRA, speculative decoding, distillation)
   - **Applications + Miscellaneous** (reasoning, vision LMs, interpretability)

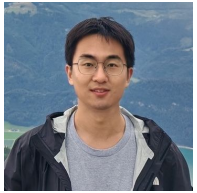
That’s it – thank you all!

Feel free to reach out if you are interested in NLP research.

NLP reading group: sign up here.

Graduate TAs

Flora Jia



Xunjian Yin

Undergraduate TAs

James Cai

Raymond Xiong

Duke Course Evaluations
https://eval-duke.evaluationkit.com/